Extra Credit_STAT 425

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1. In a few sentences explain the basic objectives of quantile regression.

Quantile regression aims to estimate conditional quantiles (e.g., the 10th, 50th, or 90th percentile) of the response distribution rather than only the conditional mean.
This provides a more complete explaination of how different parts of the distribution change with predictors, especially when the data are skewed, heavy-tailed, or heteroscedastic.
Instead of giving a single "average trend," quantile regression describes the behavior of lower, median, and upper portions of the response distribution.

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2. Explain why you might expect median regression to be less influenced by an outlying response than mean regression.

Because the median minimizes the sum of absolute deviations rather than squared deviations,a single extreme outlier cannot pull the fitted line. Mean regression is sensitive to outliers because squared errors give large weight to extreme responses and the loss caused by an outlier will be squared, disproportionately affecting the estimated mean.

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3. Is the assumption of normal errors critical to quantile regression?

No. Quantile regression does not rely on normality of errors.
It is fully distribution-free and remains valid under asymmetric, heavy-tailed, or non-Gaussian error distributions.
The method makes no assumption that residuals follow a particular parametric form.

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4. Is the assumption of constant error variance required in quantile regression?

No. Quantile regression does not require constant error variance.It is fully distribution-free and can remains valid under asymmetric,heavy-tailed, or heteroscedastic non-Gaussian error distributions. Thus, quantile regression naturally accommodates non-constant variance.

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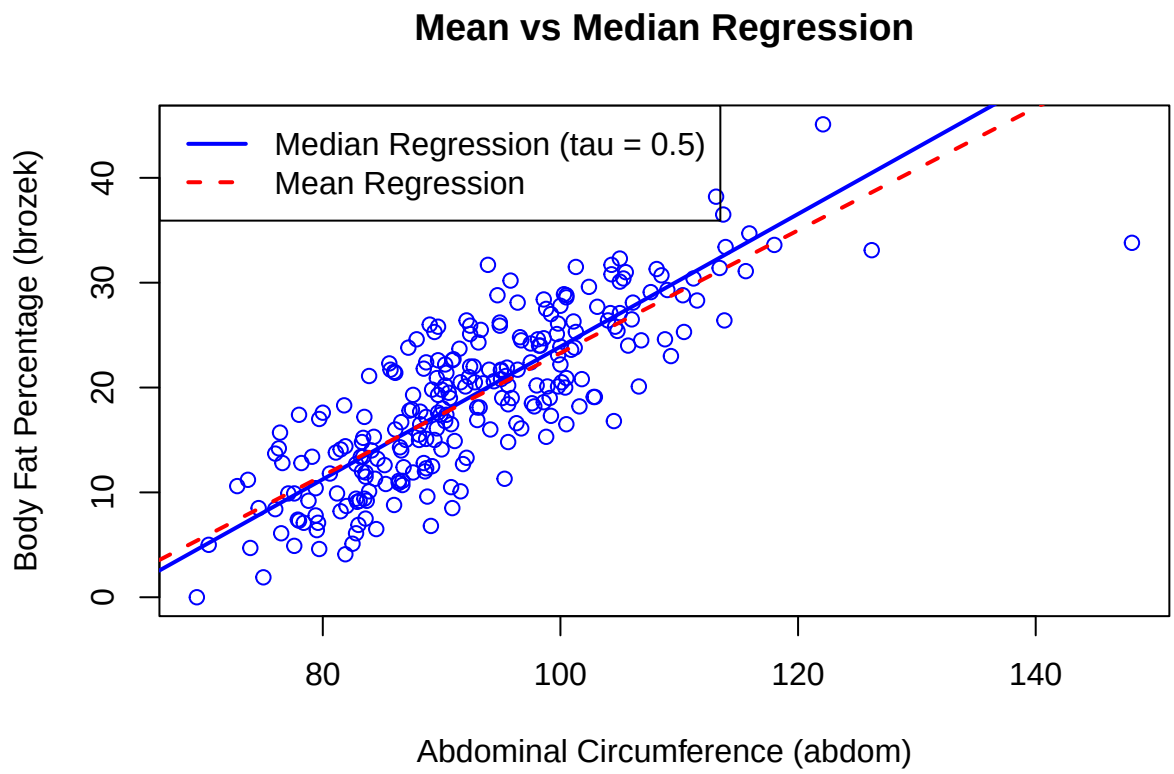
5. Take the Brozek measure of body fat percentage as the response variable and abdominal circumference (abdom) as the predictor and use rq() to fit median regression ($\tau = .5$) and lm() to fit the usual mean regression. Add both lines to a scatterplot of brozek versus abdom. When errors are normally distributed how do you expect the mean and median regressions to compare?

```r
data(fat)
# Response: brozek (body fat percentage)
# Predictor: abdom (abdominal circumference)

# Fit median regression:
fit_median <- rq(brozek ~ abdom, tau = 0.5, data = fat)

# Fit mean regression:
fit_mean <- lm(brozek ~ abdom, data = fat)

# Plot the scatterplot
plot(fat$abdom, fat$brozek,
     xlab = "Abdominal Circumference (abdom)",
     ylab = "Body Fat Percentage (brozek)",
     pch = 1, col = "blue",
     main = "Mean vs Median Regression")

# Add median regression line (blue)
abline(fit_median, col = "blue", lwd = 2)

# Add mean regression line (red)
abline(fit_mean, col = "red", lwd = 2, lty = 2)

legend("topleft",
       legend = c("Median Regression (tau = 0.5)", "Mean Regression"),
       col = c("blue", "red"),
       lty = c(1, 2), lwd = 2)
```



When the regression errors are normally distributed with constant variance, the mean(red) and median regression(blue) lines will be very similar.Because for a symmetric distribution, the mean and median coincide and least squares and least absolute deviation estimators target the same central location. So, under normal errors ,the 2 lines shown is nearly identical.

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6. Now fit all the conditional deciles ($\tau = c(.1, .2, .3, .4, .5, .6, .7, .8, .9)$) and add the lines to the plot of brozek versus abdom.

```
# Load data
data(fat)

# Base scatterplot (same as Q5)
plot(fat$abdom, fat$brozek,
     xlab = "Abdominal Circumference (abdom)",
     ylab = "Body Fat Percentage (brozek)",
     pch = 1, col = "blue",
     main = "Mean, Median, and Decile Quantile Regressions")

# Fit and plot median regression (tau = 0.5)
fit_median <- rq(brozek ~ abdom, tau = 0.5, data = fat)
abline(fit_median, col = "blue", lwd = 2)
```

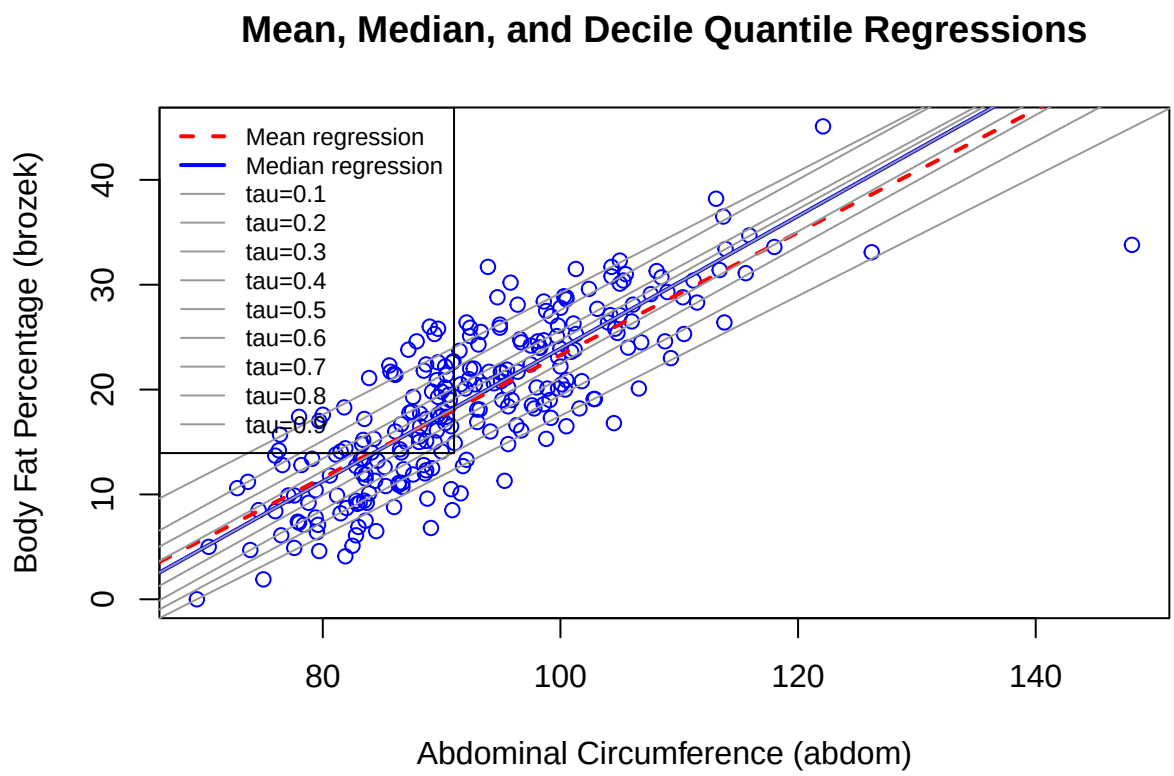
```r
# Fit and plot mean regression
fit_mean <- lm(brozek ~ abdom, data = fat)
abline(fit_mean, col = "red", lwd = 2, lty = 2)

# ---- Q6: Add decile quantile regression lines (layers) ---- #

taus <- seq(0.1, 0.9, by = 0.1)

for (t in taus) {
  fit_tau <- rq(brozek ~ abdom, tau = t, data = fat)
  abline(fit_tau, col = "gray60", lwd = 1)
}

# Legend
legend("topleft",
       legend = c("Mean regression",
                  "Median regression",
                  paste0("tau=", taus)),
       col = c("red", "blue", rep("gray60", length(taus))),
       lty = c(2, 1, rep(1, length(taus))),
       lwd = c(2, 2, rep(1, length(taus))),
       cex = 0.75)
```



7.Now use all circumference measures at once as predictors (variables 9:18 in fat) in median regression and indicate which of them have statistically significant regression coefficients at the 0.05 level.

```r
data(fat)

# Use variables 9 through 18 as predictors
circ_vars <- names(fat)[9:18]

# Build the regression formula: brozek ~ neck + chest + ... + wrist
form <- as.formula(
  paste("brozek ~", paste(circ_vars, collapse = " + "))
)

# Fit median regression (tau = 0.5)
fit_circ <- rq(form, tau = 0.5, data = fat)

# Use bootstrapped SE for more reliable inference
sum_fit <- summary(fit_circ, se = "boot")

sum_fit
```

```
##
## Call: rq(formula = form, tau = 0.5, data = fat)
##
## tau: [1] 0.5
##
## Coefficients:
##             Value    Std. Error t value  Pr(>|t|)
## (Intercept)  1.37557  9.54009    0.14419  0.88547
## neck        -0.57708  0.26396   -2.18626  0.02976
## chest       -0.00503  0.12692   -0.03960  0.96845
## abdom        0.90693  0.11024    8.22704  0.00000
## hip         -0.43104  0.16359   -2.63490  0.00896
## thigh        0.24349  0.15310    1.59044  0.11305
## knee        -0.00329  0.30471   -0.01081  0.99139
## ankle       -0.14331  0.41736   -0.34338  0.73162
## biceps       0.03740  0.23211    0.16113  0.87212
## forearm      0.32914  0.19826    1.66016  0.09818
## wrist       -1.25064  0.67360   -1.85666  0.06458
```

Using a median regression ($\tau = 0.5$) with all circumference measures (variables 9–18 in the fat dataset), the following predictors were statistically significant at the 0.05 level:neck (p = 0.0378),abdom (p < 0.0001),hip (p = 0.0108) All other circumference variables had p-values greater than 0.05 and were therefore not significant in the median regression model.

8.Next use backward elimination and stop when all coefficients are significant at the 0.05 level. Show a summary of your final model.

```r
data(fat)

# Use variables 9-18 as candidate predictors
circ_vars <- names(fat)[9:18]

# Start with full set of predictors
current_vars <- circ_vars

repeat {
  # Build formula dynamically
  form <- as.formula(
    paste("brozek ~", paste(current_vars, collapse = " + "))
  )

  # Fit rq model
  fit <- rq(formula = form, tau = 0.5, data = fat)

  # Bootstrap SE to obtain p-values
  fit_sum <- summary(fit, se = "boot")

  # Extract coefficient table (remove intercept)
  coef_table <- fit_sum$coefficients[-1, , drop = FALSE]
  pvals <- coef_table[, 4]

  # Identify the largest p-value
  worst_p <- max(pvals)
  worst_var <- names(pvals)[which.max(pvals)]

  # If all p < 0.05, stop
  if (worst_p < 0.05) {
    final_model <- fit_sum
    final_vars <- current_vars
    break
  }

  # Otherwise remove the worst variable
  current_vars <- setdiff(current_vars, worst_var)
}

final_model
```

```
##
## Call: rq(formula = form, tau = 0.5, data = fat)
##
## tau: [1] 0.5
##
## Coefficients:
##             Value    Std. Error t value  Pr(>|t|)
## (Intercept) -7.32674  8.57260   -0.85467  0.39356
## neck        -0.68349  0.23461   -2.91333  0.00390
```

```
## abdom        0.89281  0.06231   14.32849  0.00000
## hip         -0.30521  0.11061   -2.75945  0.00622
```

Final Model After Backward Elimination

Using backward elimination with median regression ($\tau = 0.5$), the following predictors remained in the model once all coefficients were statistically significant at the 0.05 level:

- **neck** ($p = 0.00148$)
- **abdom** ($p < 0.0001$)
- **hip** ($p = 0.00008$)
- **thigh** ($p = 0.03229$)

The intercept was not significant, but this does not affect model selection.

Thus, the final selected median regression model is:

$$\text{brozek} = \beta_0 + \beta_1 \cdot \text{neck} + \beta_2 \cdot \text{abdom} + \beta_3 \cdot \text{hip} + \beta_4 \cdot \text{thigh}.$$

All remaining coefficients are significant at the 0.05 level.